Melanoma Detection Model Case Study Rubric

DS 4002 – Harry Shield

Submission format: Upload link to GitHub repository on UVA Canvas

Individual Assignment

Why am I doing this?

Following this case study challenges your skills as a data scientist and prepares you for future coursework and career opportunities. This project specifically tests your coding abilities and your experience working with visual image models. You will practice building predictive models, analyzing data, and becoming more comfortable with different modeling approaches.

What am I going to do?

The GitHub repository for this case study can be found here.

You will obtain the data from the ISIC Skin Lesion Daset, available here. Cleaned metadata for the model also present in materials.

After obtaining and cleaning the data (or using precleaned data), you will create multiple plots to explore the link between diagnosis and age, sex, and lesion location. Once you understand the trends you will select two pre-trained visual PyTorch models, specifically ResNet18 and DenseNet121 to compare.

You will:

- Train both models
- Evaluate their F1 and Recall metrics and compare their performance; make sure calculations are performed with data separate from the training set

Your final deliverables should include:

- The original and cleaned datasets you used, with data dictionary
- Plots showing link of diagnosis with age, sex, and lesion location
- Well documented and commented code
- A CSV with model evaluation metrics
- A GitHub repository containing all project materials

Tips for success:

- Understanding how ResNet18 and DenseNet121 work, as well as how F1 and Recall are calculated

- Organize GitHub repository using folders
- Be comfortable using Python, as this will be your main coding language

How will I know I have succeeded? You will meet expectation on this case study when you successfully follow and complete the criteria in the rubric below:

Formatting	• One GitHub Repository (submitted via link on canvas)
	• Organized with these folders
	  ◦ A README.md file (which auto displays)
	  ◦ A LICENSE.md file (use MIT as default)
	  ◦ A SCRIPTS folder
	  ◦ A DATA folder
	  ◦ AN OUTPUT folder

README.md	• Goal: This file serves as an orientation to everyone who comes to your repository, it should enable them to get their bearings.
	• Use markdown headers to divide content.
	• Make an H2 (##) section explaining the contents of the repository
	• ***Section 1: Software and platform section***
	  ◦ The type(s) of software you used for the project.
	  ◦ The names of any add-on packages that need to be installed
	  ◦ The platform (e.g., Windows, Mac, or Linux) you used.
	
	• ***Section 2: A Map of your documentation.***
	
	In this section, you should provide an outline or tree illustrating the hierarchy of folders and subfolders contained in your Project Folder
	
	• ***Section 3: Instructions for reproducing your results.***
	
	In this section, you should give explicit step-by-step instructions to reproduce the Results of your study.

LICENSE.md	• Goal: This file explains to a visitor the terms under which they may use and cite your repository. • Select the MIT license is appropriate.
SCRIPTS Folder	• Goal: This folder contains all the source code for your project. • Include scripts for each model and the code for making the plots. Try to name each script according to what it completes
OUTPUT Folder	• Goal: This folder contains all of the output generated by your project, e.g. figures, tables, etc. • Include plots from the EDA and the CSV comparing F1 and recall performance
DATA Folder	• Goal: This folder contains all of the data for this project. • Include original and cleaned data
References	• All references should be listed at the end of the document • Use IEEE Documentation style (link)